

Econ 5260: Macroeconomic Theory **Final Exam**

May 22th, 2026

The exam lasts 3 hours. Time will be a factor, so be careful. Read the exam questions carefully before you start.

Part A: Discussion Questions Please explain your answer briefly. Each question is worth 10 marks.

A1: From Lecture 10, we know that Woodford(2001a) shows that deviation of the expected discounted utility from its steady state can be approximated by an equation which is similar to the standard loss function. Explain intuitively why inflation variability enters into that loss function in Woodford's model.

A2: An exogenous policy rule-one that does not respond to the endogenous variables such as the output gap x and the inflation π may introduce the possibility of multiple equilibria. Is this statement true or false. Please explain.

Part B: Short Answer Questions (30 marks)

B1: Nakamura and Steinsson (QJE 2008) define T as the duration of a general price spell and the hazard of a price change at time t as the probability that the price will change after t periods given that it has survived for t periods. Please write out the hazard function and explain what should be the slope of the hazard function under the TDP and SDP model. What is the shape of the hazard function implied by the data found in Nakamura and Steinsson (QJE 2008)? Discuss if it is consistent with the sticky price models we studied in class.

Ph.D students only

B2: Analyze and explain briefly the similarities and difference between the Sticky price Phillips Curve and sticky information Phillips Curve. Does the Sticky information Phillips curve do better in explaining inflation persistence?

Part C: Long questions

C1: (25 marks)

Suppose the central bank attempts to minimize the following quadratic loss function

$$L = E_t \sum_{i=0}^{\infty} \beta^i (\pi_{t+i}^2 + \lambda x_{t+i}^2) \quad (0.1)$$

subject to the expectation augmented IS curve (all variables expressed as percentage deviations around a zero inflation steady state):

$$x_t = E_t(x_{t+1}) - \frac{1}{\sigma}(i_t - E_t\pi_{t+1}) + u_t \quad (0.2)$$

where x_t is the gap between output and the flexible-price equilibrium output level, and u_t is the demand shock (taste shock, government demand shock, and fluctuation of flexible price-equilibrium level).

and the AS curve (again, all variables expressed as percentage deviations around a zero inflation steady state):

$$\pi_t = \beta E_t\pi_{t+1} + \kappa x_t + e_t \quad (0.3)$$

where e_t is called an inflation shock, a cost-push shock, or a price shock and satisfies $e_t = \rho e_{t-1} + \epsilon_t$, where ϵ_t is a white noise.

(a) First, assume that the central bank is able to precommit a path for current and future inflation and the output gap (choose i_{t+i} , π_{t+i} , and x_{t+i}) to minimize

the loss function (0.1) subject to the IS curve and the AS curve. Find the optimal commitment policy. Is there dynamic inefficiency?

(b) Now consider a **timeless perspective approach** to precommitment and let

$$\pi_{t+i} = -\frac{\lambda}{\kappa}(x_{t+i} - x_{t+i-1}) \quad (0.4)$$

for all $i \geq 0$. Substitute 0.4 to 0.3. And then assume that $x_t = a_x x_{t-1} + b_x e_t$ and solve for the equilibrium inflation under the timeless perspective policy. Explain why, under commitment, the central bank promises a deflation in the period after a positive cost shock (assume the cost shock is serially uncorrelated).

(c) Now assume that the central bank operates with discretion. That is, it acts each period to minimize the loss function given by 0.1. Since the action at t does not bind at any future periods, the central bank is unable to affect private sector's expectation about future inflation. Guessing a solution of the form $x_t = \delta e_t$ and then δ and the equilibrium inflation under discretion.

(d) Compare your answer to b) and c). Is the response of inflation to cost shock the same? Is there a stabilization bias? Is there inflation bias? Explain your result intuitively.

e) Now suppose the central bank can commit to a rule of the form, $x_t = b_x e_t$, then

$$\pi_t = \beta E_t \pi_{t+1} + \kappa b_x e_t + e_t \quad (0.5)$$

Find out the equilibrium inflation when the central bank commit to a rule. If $0 < \rho < 1$, explain if there is stabilization bias and if this stabilization bias is higher or lower than that under the discretionary monetary policy.

C2:(25 marks)

Consider a limited participation model. There is a continuum of perfectly competitive, final goods firms that produce consumption goods using intermediate goods. Their production function are

$$y = \int_0^1 [y(i)^\lambda di]^\frac{1}{\lambda} \quad (0.6)$$

Suppose the price of final goods and the price of intermediate goods i are P and $p(i)$, respectively.

(a) Set up the optimization problem of final goods firm and derive the demand for intermediate good i .

Intermediate good firms are monopolistic competitive in the product market and competitors in the market for labor. They set price for their goods and are required to supply whatever final goods producer demand at those prices. These firms need to disburse their wage bill before their realized their revenue, so in each period, they borrow an amount L_t from financial intermediaries and use that to finance the wage bill. They pay principle and interest rate, at gross interest rate R_t , on their loan at the end of the period.

(b) Set up the optimization problem of intermediate good firms and solve for the optimal price setting equation.

Financial intermediaries behave competitively. They receive D_t in deposit from household, and $X_t = M_{t+1} - M_t$ on the household's behalf from the central bank, and then lend the loan out to firms. In the end of the period, the intermediary pays $R_t D_t$ to households for their deposits, and distribute $R_t X_t$ to the households in the form of profits. It is assumed that the representative financial intermediary maximizes its cash flow.

(c) Set up the optimization problem for FI and solve for the first order condition when $R_t > 1$.

The representative household choose sequences of consumption and leisure to maximizes his expected utility $E_0 \sum_{t=0}^{\infty} \beta^t U(c_t, l_t)$ where

$$U = \frac{1}{1-\sigma} c^{1-\sigma} - \frac{1}{1+\psi} (1-l)^{1+\psi} \quad (0.7)$$

Households enter the period with currency holding M_t and make deposits D_t with financial intermediaries;

$$0 \leq D_t \leq M_t \quad (0.8)$$

They also face the following CIA constraint on goods markets;

$$P_t c_t \leq W_t(1-l_t) + M_t - D_t \quad (0.9)$$

and their end of period wealth is given by

$$M_{t+1} = W_t(1-l_t) + M_t - D_t - P_t c_t + R_t(D_t + X_t) + \Pi_t \quad (0.10)$$

where Π_t are the profits of intermediate goods firms.

(d) Consider a stationary equilibrium where $d = D/M$, $p = P/M$, and $w = W/M$, and $x = X/M$. Show that the nominal interest rate R decreases in x and consumption/output is increasing in x .

For Ph.D Students Only

(e) If x is i.i.d., with support $\{x_l, x_h\}$ and probability distribution $\{\frac{1}{2}, \frac{1}{2}\}$, solve for d by using households' Euler equation.